A2 · Unit 4 · WBI14 Energy, Environment, Microbiology and Immunity

Photosynthesis

Photosynthesis is how green plants, algae and some bacteria turn light energy into chemical energy stored in sugars. Almost every food chain starts here, and the oxygen in the air is a by-product of the process. This page follows the energy from the moment light strikes a pigment molecule to the sugars made in the stroma, and explains why the rate of the process rises and falls.

SPEC 5.1 Photosynthesis: the overall reaction

WHAT THIS SPEC POINT ASKS

Understand photosynthesis as using light energy to split the strong bonds in water, storing the hydrogen in a fuel (glucose) by combining it with carbon dioxide, and releasing oxygen.

The big picture

- Plants, algae and some bacteria are **autotrophs**: they make their own organic molecules from simple inorganic ones (CO_2 and water), using **light energy**. Animals and fungi are **heterotrophs**: they get organic molecules by eating or absorbing them.
- Photosynthesis turns **light energy** into **chemical energy** stored in the bonds of glucose. Respiration later releases that energy.

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (light energy, chlorophyll)

What happens to each reactant

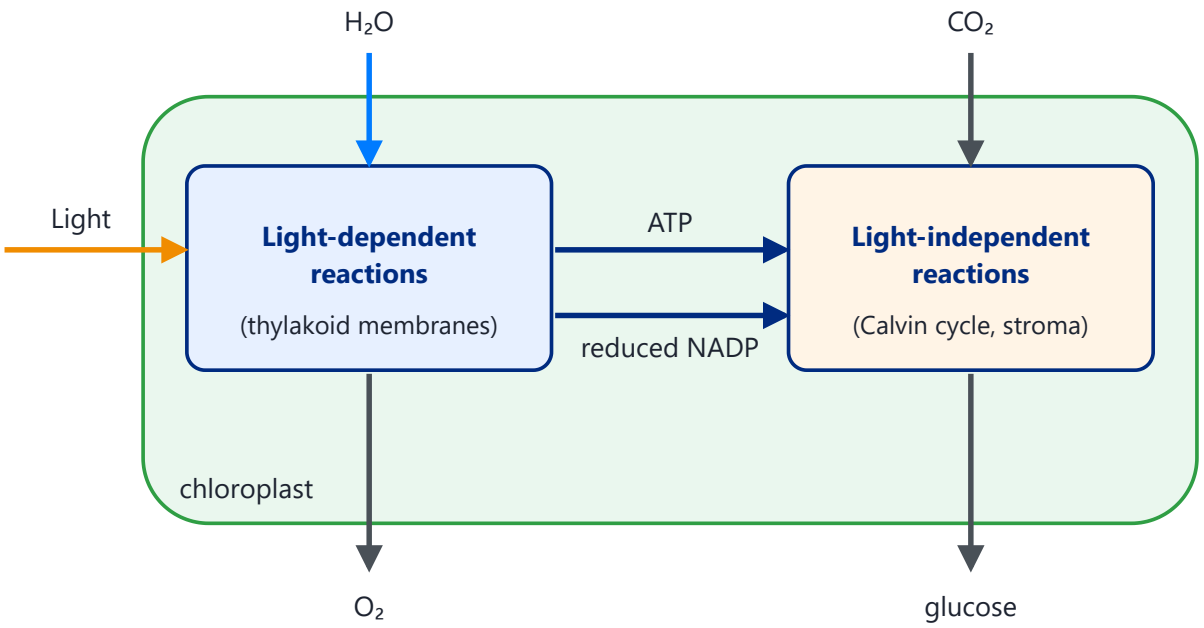
- **Water**: the O–H bonds in water are strong. Light energy is needed to break them (**photolysis**). This gives hydrogen (as H^+ ions and electrons) and **oxygen**, which is released as a **waste product**.
- **Hydrogen** is the part worth keeping. It is passed on (via reduced NADP) and combined with CO_2 to make glucose. Glucose is the "fuel" that stores the hydrogen.
- **Carbon dioxide** supplies the **carbon** and the **oxygen** in glucose.

Atoms in glucose	Come from
Carbon	carbon dioxide
Oxygen	carbon dioxide
Hydrogen	water

Atoms in glucose	Come from
The O_2 released	water

Two stages (details in 5.3 and 5.4)

Stage	Where	Needs	Makes
Light-dependent reactions	thylakoid membranes	light, water, ADP + Pi, NADP	ATP, reduced NADP, O_2 (waste)
Light-independent reactions (Calvin cycle)	stroma	CO_2, ATP, reduced NADP	GALP → glucose and other organic molecules; ADP, Pi and NADP go back to the thylakoids



The two stages are linked: ATP and reduced NADP flow from the light-dependent reactions to the light-independent reactions.

KEY IDEA

Photosynthesis is a **redox** process. Water is **oxidised** (loses electrons and hydrogen). Carbon dioxide is **reduced** (gains hydrogen). Light supplies the energy to push the hydrogen "uphill" into glucose.

WORKED EXAMPLE

Worked example: energy from bond energies

A question gives the O–H bond energy as 463 kJ mol^{-1} and says one O–H bond breaks in each water molecule in the equation $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$.

Energy needed = number of molecules × bonds broken per molecule × bond energy = 6 × 1 × 463 = **2778 kJ** per mole of glucose.

The skill is reading the stem: which bonds, how many per molecule, how many molecules. Bonds in glucose itself are C–H, C–O, C–C and O–H.

MARK-SCHEME WORDING

- Sources of atoms: "C and O from carbon dioxide", "H from water". **Not** "O (in glucose) from water".
- Name the **molecules** (carbon dioxide, water), not just the elements.
- Why not all the sunlight is used: "not all wavelengths can be absorbed", "**green** light is reflected", "light is transmitted through the leaf". Shading and competition are ignored.
- Glucose is made in the **stroma**.

Often asked as: where the C, H and O in glucose come from; which stage uses CO_2 and which releases O_2 (MCQ); bond-energy calculations; why plants capture only a small share of the light energy.

COMMON MISTAKE

Saying the oxygen in glucose comes from water. The O_2 **released** comes from water; the O **in glucose** comes from CO_2.

"Light reflected" on its own: say **green** light is reflected.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Bond-energy sums: pick the right bonds and multiply by the number of molecules. Candidates often missed which bond types (C–H, C–O, C–C) were needed to reach a total (WBI14 Oct 2020).
- Losses of light energy before photosynthesis even starts: reflected (green), transmitted, wrong wavelength, hits non-photosynthetic parts (WBI14 Oct 2024).

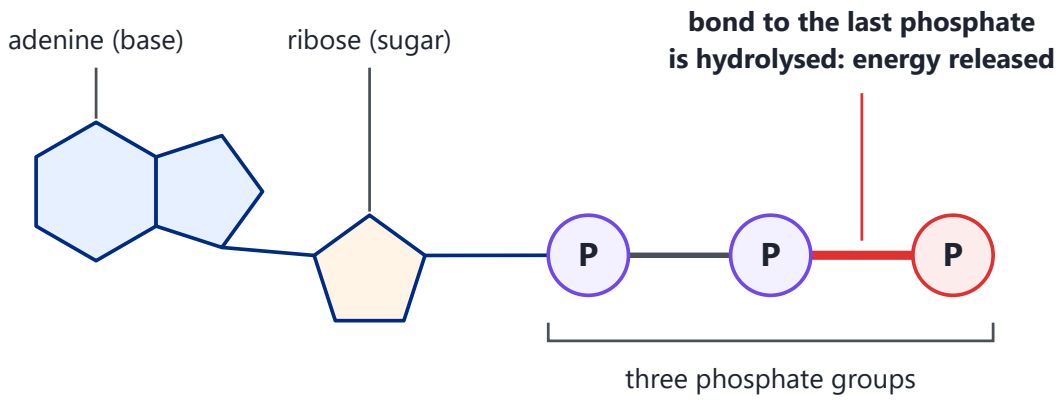
SPEC 5.2 ATP: photophosphorylation and hydrolysis

WHAT THIS SPEC POINT ASKS

Understand that making ATP from ADP (photophosphorylation, in photosynthesis) needs energy, and that hydrolysing ATP gives an immediate supply of energy for biological processes.

What ATP is

- **ATP (adenosine triphosphate)** is a phosphorylated nucleotide: the base **adenine**, the pentose sugar **ribose**, and **three phosphate groups**.
- Removing one phosphate gives **ADP** (adenosine diphosphate) and **inorganic phosphate (Pi)**.



ATP is adenine and ribose attached to three phosphate groups; hydrolysis of the last phosphate releases energy.

Making ATP needs energy

ADP + Pi + energy → ATP + H_2O

- Adding a phosphate to ADP is **phosphorylation**, a condensation reaction catalysed by **ATP synthase**.
- The energy can come from:
  - light, in the chloroplast: **photophosphorylation** (5.3);
  - the oxidation of respiratory substrates, in the mitochondrion: oxidative phosphorylation (Topic 7).
- In both, the energy first builds a **proton (H^+) gradient** across a membrane. H^+ flow back through ATP synthase, and that flow drives phosphorylation (**chemiosmosis**).

Hydrolysing ATP releases energy

ATP + H_2O → ADP + Pi + energy

- Catalysed by **ATP hydrolase** (ATPase).
- Releases a **small, manageable** amount of energy (about 30 kJ mol^{-1}), so little is wasted as heat.
- It is **one reaction, one enzyme**, so energy is released **immediately**, exactly where it is needed. (Releasing energy from glucose takes dozens of steps.)
- ATP is **small and soluble**, so it moves easily around the cell.
- The released phosphate can be attached to another molecule (phosphorylating it), making it more reactive. This **couples** ATP hydrolysis to energy-requiring reactions.
- ATP is **not a store**: it is used within seconds and constantly remade from ADP + Pi. Glucose, starch and lipids are the stores.

Uses in plants: the **Calvin cycle** (reducing GP and regenerating RuBP), active transport of mineral ions into roots, protein synthesis, phloem loading.

KEY IDEA

Plants trap light energy, but light energy cannot drive reactions directly. It is first transferred to **ATP** (and reduced NADP), which then supply the energy for the Calvin cycle.

MARK-SCHEME WORDING

- "ATP is the {usable source of energy / energy currency}"; "light energy cannot be used (directly)"; "ATP is needed in the Calvin cycle to convert GP into GALP".
- Products of hydrolysis: "ADP and (inorganic) phosphate / Pi". **Not** "P" or "phosphorus". "Dephosphorylation" is accepted.
- Phosphorylation: "hydrogen ions pass through ATP synthase, releasing energy for phosphorylation of ADP".
- MCQ logic: ADP is **phosphorylated** and this **needs** energy; ATP is **hydrolysed** and this **releases** energy.

Often asked as: why plants need ATP even though they trap light; products of ATP hydrolysis; fill-in blanks on a photophosphorylation diagram (ADP, Pi, ATP); MCQs.

COMMON MISTAKE

"ATP makes energy" or "ATP is energy". Energy can't be made. ATP is a **source** or **carrier** of energy.

Writing "P" for phosphate. If unsure of the symbol, write the word **phosphate**.

Answering "how light is converted into ATP" when the question asks **why** plants need ATP.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **ATP synthase** and **chemiosmosis** appear in mark schemes for the light-dependent reactions (WBI14 Jan 2021).
- Glucose made in photosynthesis is respired to give the ATP needed for protein synthesis (WBI14 Jan 2020).

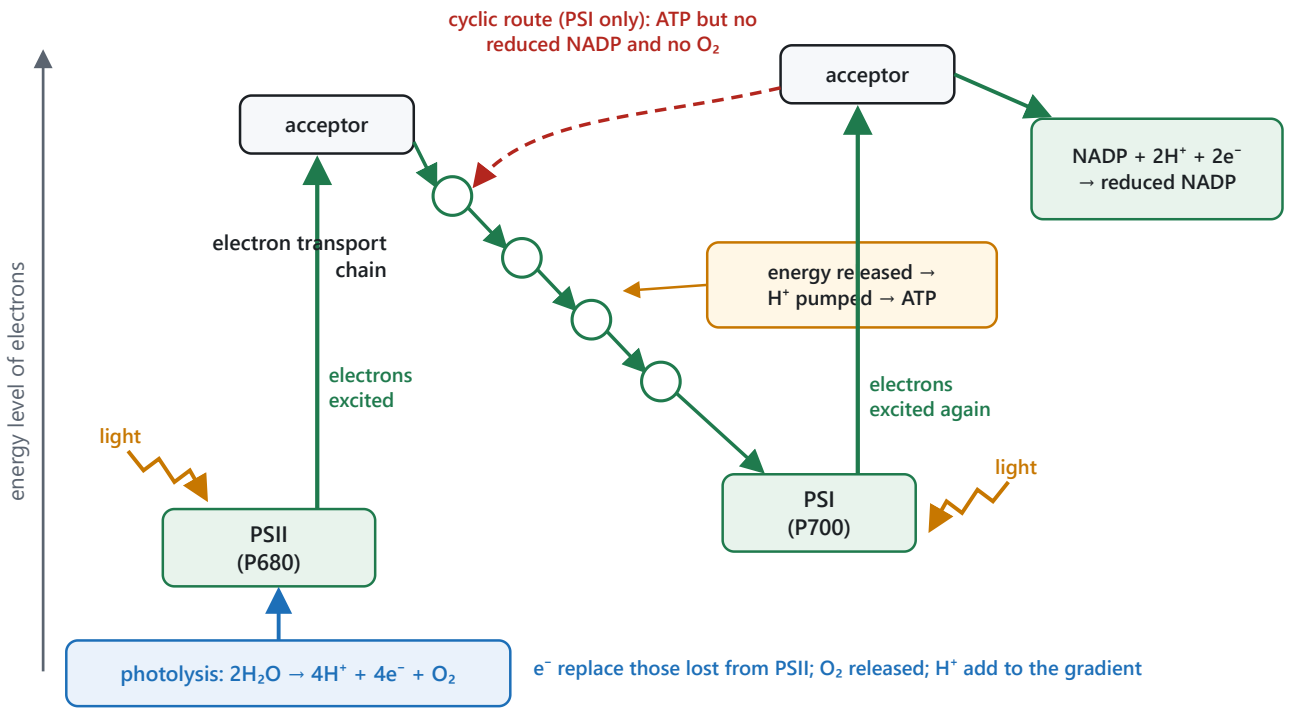
SPEC 5.3 The light-dependent reactions

WHAT THIS SPEC POINT ASKS

Understand how light energy excites electrons in chlorophyll, how those electrons drive ATP production and NADP reduction in cyclic and non-cyclic photophosphorylation, and how photolysis of water produces oxygen.

Where and what

- On the **thylakoid membranes** of the chloroplast.
- Chlorophyll and accessory pigments sit in protein complexes called **photosystems**. Each has a **reaction centre** chlorophyll a molecule. **PSII** absorbs best at 680 nm (P680); **PSI** at 700 nm (P700). (They are numbered in order of discovery, not the order electrons pass through them.)
- Products: **ATP**, **reduced NADP**, and **oxygen** (waste).



Non-cyclic route (solid green): water → PSII → electron transport chain → PSI → NADP. Cyclic route (dashed red): electrons from PSI go back down the chain to PSI, making ATP only.

Non-cyclic photophosphorylation, step by step

1. Light is **absorbed** by chlorophyll in **PSII**. Its energy **excites electrons** to a higher energy level, and they leave the chlorophyll (the chlorophyll is oxidised).
2. The electrons are taken up by an **electron acceptor** and passed along a chain of **electron carriers** (the **electron transport chain**) in a series of redox reactions.
3. As they move down the chain, the electrons **lose energy**. This energy pumps **H^+ from the stroma into the thylakoid space**.
4. H^+ build up in the thylakoid space (a **proton gradient**). They diffuse back into the stroma through **ATP synthase**, which uses the energy to **phosphorylate ADP → ATP** (**chemiosmosis**).
5. The electrons reach **PSI**. Light absorbed by PSI **excites them again**.
6. The electrons pass to **NADP**. Together with **H^+** it becomes **reduced NADP**.
7. **Photolysis**: PSII has lost electrons, so it must get them back. Light energy splits water:

$2H_2O \rightarrow 4H^+ + 4e^- + O_2$

- The **electrons** replace those lost from PSII.
- The **H^+** add to the proton gradient and are used to reduce NADP.
- The **oxygen** diffuses out of the leaf as a by-product (or is used in respiration).

The electron path is one-way, from water to NADP, which is why it is called **non-cyclic**.

Following the electrons through non-cyclic flow

1. Light absorbed by the antenna of photosystem II excites electrons in P680. They leave the chlorophyll and pass to an electron acceptor.
2. Photolysis of water replaces the lost electrons, releasing oxygen and adding hydrogen ions to the lumen.

   Without this supply of electrons, P680 could not absorb light again and the flow would stop.
3. The electrons are passed along a chain of electron carriers in the thylakoid membrane in a series of redox reactions, losing energy at each step.
4. Some of that energy is used to pump hydrogen ions from the stroma across the thylakoid membrane into the lumen.

   Moving ions against their concentration gradient needs energy, which comes from the electrons.
5. Hydrogen ions are now much more concentrated in the lumen than in the stroma. They can only diffuse back through channels in ATP synthase.
6. The flow of hydrogen ions through ATP synthase provides the energy to join ADP and Pi into ATP. Making ATP this way is chemiosmosis, and because light drives it, the process is photophosphorylation.
7. The electrons reach photosystem I, where light excites them again from P700 to a higher energy level, and they pass to a second electron acceptor.
8. The enzyme NADP reductase uses these electrons, together with hydrogen ions from the stroma, to reduce NADP to reduced NADP. Reduced NADP carries hydrogen to the Calvin cycle.

Cyclic photophosphorylation

- Uses **PSI only**. Excited electrons from PSI pass to the electron transport chain and then **return to PSI**.
- Their energy still pumps H^+, so **ATP is made**.
- **No** reduced NADP, **no** photolysis, **no** oxygen.

	Non-cyclic	Cyclic
Photosystems	PSII and PSI	PSI only
Electron source	water (photolysis)	PSI (electrons go round)
Final electron acceptor	NADP	PSI

	Non-cyclic	Cyclic
ATP	yes	yes
Reduced NADP	yes	no
O_2 released	yes	no

EXAM TIP

For "describe how light energy is used to make ATP", cover **both** routes: ATP is made in cyclic **and** non-cyclic photophosphorylation. Very few candidates say this, and the 4-mark version needs it.

MARK-SCHEME WORDING

- "Light is **absorbed** by {chlorophyll / photosystems}, which **excites electrons**" (or "releases high-energy electrons").
- "Electrons are passed along a series of electron carriers, releasing energy to phosphorylate ADP into ATP."
- Photolysis "{replaces electrons lost by chlorophyll / provides protons for formation of reduced NADP}".
- Purpose of the light-dependent reactions: "ATP and reduced NADP for the Calvin cycle".
- The product made only in non-cyclic is **reduced NADP**. NADPH is accepted; NADP, NADH or reduced NAD are not.
- Location: thylakoid **membranes**; H^+ accumulate in the **thylakoid space**.
- Magnesium: "a component of chlorophyll, which absorbs light".

Often asked as: describe how light leads to ATP (4 marks); role of photolysis; role of chlorophyll; which product is only in non-cyclic; location MCQs; why a shortage of water or Mg^{2+} slows photosynthesis.

COMMON MISTAKE

Light "hits" or "strikes" chlorophyll. The word that scores is **absorbed**. Examiners have flagged this in report after report.

Saying NADP is "made" in the light-dependent reactions. It is **reduced**.

Mixing up cyclic and non-cyclic: oxygen and reduced NADP come only from **non-cyclic**.

Saying the light-dependent reactions happen in the stroma, or "in the grana" without the membrane.

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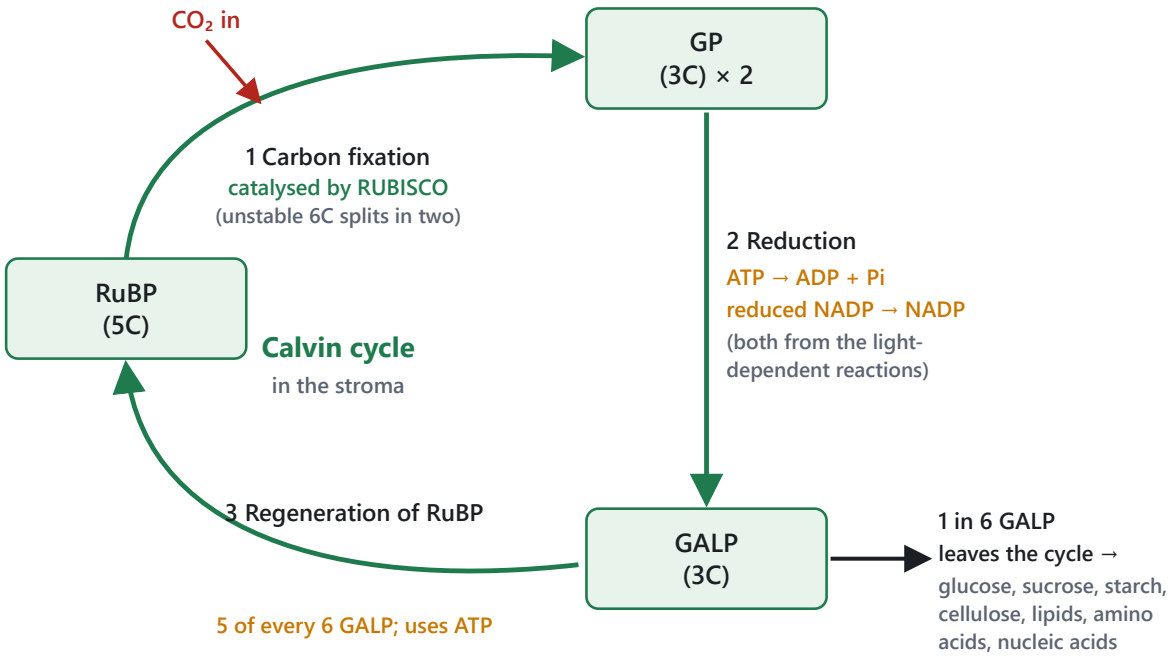
- **Mg^{2+}** is part of the chlorophyll molecule; Mg^{2+} uptake by roots needs active transport, so ATP and oxygen for aerobic respiration (WBI14 Jun 2023).
- **Cyanobacteria** have no chloroplasts: their light-dependent reactions happen on internal membranes (WBI14 Jan 2023).
- Less water reaching the leaf means less photolysis, so fewer electrons and H^+ for the light-dependent reactions (WBI14 Jun 2024).
- Why cyclic exists: the Calvin cycle uses 3 ATP for every 2 reduced NADP. Non-cyclic alone gives too little ATP, so cyclic tops it up.

SPEC 5.4 The light-independent reactions (Calvin cycle)

WHAT THIS SPEC POINT ASKS

(i) Understand the light-independent reactions as the reduction of CO_2 using ATP and reduced NADP: carbon fixation in the Calvin cycle and the roles of GP, GALP, RuBP and RUBISCO. (ii) Know that the simple sugars made are used for respiration and to build polysaccharides, amino acids, proteins, lipids and nucleic acids.

The cycle in three steps (in the stroma)



Carbon fixation, reduction and regeneration. One GALP in every six leaves the cycle to make useful molecules.

1. **Carbon fixation.** CO_2 diffuses into the stroma and combines with **RuBP** (ribulose bisphosphate, a **5-carbon** compound). The enzyme is **RUBISCO**. The 6-carbon product is unstable and splits at once into **two molecules of GP** (glycerate 3-phosphate, **3-carbon**).
2. **Reduction.** GP is reduced to **GALP** (glyceraldehyde 3-phosphate, 3-carbon; some books call it triose phosphate, TP, or G3P).
   - **Reduced NADP** supplies the hydrogen (it is oxidised back to NADP).
   - **ATP** supplies the energy (hydrolysed to ADP + Pi).
   - The NADP, ADP and Pi return to the thylakoids to be used again.
3. **Regeneration of RuBP.** Most GALP (**5 in every 6**) is rearranged back into RuBP. This uses **more ATP**. Without regeneration the cell would run out of RuBP and **carbon fixation would stop**.

One turn of the Calvin cycle

1. Carbon dioxide diffuses into the leaf through the stomata and into the stroma. The enzyme RUBISCO catalyses its combination with ribulose bisphosphate (RuBP), a 5-carbon compound. This is carbon fixation.
2. The 6-carbon product is unstable and immediately splits into two molecules of glycerate 3-phosphate (GP), a 3-carbon compound.
3. GP is reduced to glyceraldehyde 3-phosphate (GALP). Reduced NADP supplies the hydrogen and ATP supplies the energy, one of each for every GP.

   This is the reduction step. Adding hydrogen stores chemical energy in the product.
4. Of every 12 GALP formed, 10 are used to regenerate 6 RuBP, which uses a further 6 ATP. The other 2 GALP leave the cycle and can form one glucose or other organic compounds.
5. Fixing 6 CO_2 therefore uses 18 ATP (12 for reduction and 6 for regeneration) and 12 reduced NADP. The ADP, Pi and NADP return to the thylakoids to be recharged.

   Without a constant supply of ATP and reduced NADP, RuBP cannot be regenerated and carbon fixation soon stops.

Counting carbons

WORKED EXAMPLE

Worked example: six turns of the cycle

6 CO_2 + 6 RuBP (6 × 5C = 30C) → 12 GP (36C)

12 GP → 12 GALP, using **12 ATP** and **12 reduced NADP**

10 GALP (30C) → 6 RuBP (30C), using **6 ATP**

2 GALP (6C) left over → **1 glucose** (6C)

Totals per glucose: **18 ATP** and **12 reduced NADP**, so ATP : reduced NADP = 18 : 12 = **1.5 : 1**.

Per CO_2 fixed: 3 ATP and 2 reduced NADP.

What happens if light or CO_2 changes

Change	GP	RuBP	Why
Light reduced (or switched off)	rises	falls	less ATP and reduced NADP, so GP isn't reduced to GALP and builds up; RuBP is still used in fixation but is not regenerated
CO_2 reduced	falls	rises	less fixation, so less GP made; RuBP is still regenerated from GALP but not used up
Temperature lowered	both processes slow		RUBISCO and other Calvin cycle enzymes have less kinetic energy, so fewer enzyme–substrate complexes form

The Calvin cycle does not need light directly, but it **stops within seconds in the dark** because it runs out of ATP and reduced NADP. Don't call it the "dark reaction".

The model in the interactive section at the bottom of the page lets you make these changes and watch GP and RuBP respond.

What the products are used for

Product	How it is made	Extra element needed
Glucose (hexose)	two GALP join	none
Sucrose (transport in phloem)	glucose + fructose	none
Starch (storage), **cellulose** (cell walls)	polymers of glucose	none
Lipids	glycerol from GALP; fatty acids from GALP via acetyl coenzyme A	none
Amino acids → proteins	from sugars + nitrogen; joined by **peptide bonds** in condensation reactions at ribosomes	**nitrogen** (from nitrate); some need **sulfur** (from sulfate)

Product	How it is made	Extra element needed
Nucleotides → nucleic acids (and ATP)	sugar + base + phosphate	**nitrogen** (bases) and **phosphorus** (from phosphate)

- Some glucose is **respired** to make the ATP these syntheses need.
- Animals, fungi and decomposers get these molecules by eating plants (or eating what ate them), so the sugars made here fuel almost every food chain.

EXAM TIP

"How does GALP become protein / root biomass?" wants a short chain, not the whole Calvin cycle: GALP → glucose → (with fructose) **sucrose** → **transported in phloem** → converted to amino acids **using nitrates** → joined by peptide bonds → protein; some glucose respired for ATP.

MARK-SCHEME WORDING

- Fixation: "carbon dioxide {binds to / reacts with} RuBP", "catalysed by RUBISCO", "forming GP".
- Regeneration: "(otherwise) the cell will run out of RuBP", so "no carbon fixation".
- "GALP converted into {hexose / glucose}"; "amino acids made from glucose and nitrates"; "amino acids joined by peptide bonds / condensation"; "glucose respired to produce ATP".
- Why glucose alone can't make amino acids: "amino acids contain **nitrogen**", "some contain **sulfur**", "nitrogen from nitrates, sulfur from sulfates". "Amino acids have an amino group" alone is too vague.
- Tick grids: protein needs nitrate; nucleic acid needs nitrate **and** phosphate; triglyceride needs neither.
- Transport: "glucose and fructose used to produce sucrose", "sucrose / amino acids moved in the phloem". **Not** "glucose transported".
- GP and GALP have accepted alternative names (triose phosphate, G3P). Carbon numbers are not required but help.

Often asked as: describe carbon fixation; why most GALP stays in the cycle; ATP : reduced NADP ratio; how GALP becomes protein, root biomass or another product; why a plant needs nitrate or sulfate; temperature and RUBISCO.

COMMON MISTAKE

Writing out the whole Calvin cycle when only "GALP plus nitrate → amino acids" was wanted.

Forgetting **nitrate** when explaining how plants make proteins.

Saying glucose is transported in the phloem. It is **sucrose**.

Saying the Calvin cycle "happens at night".

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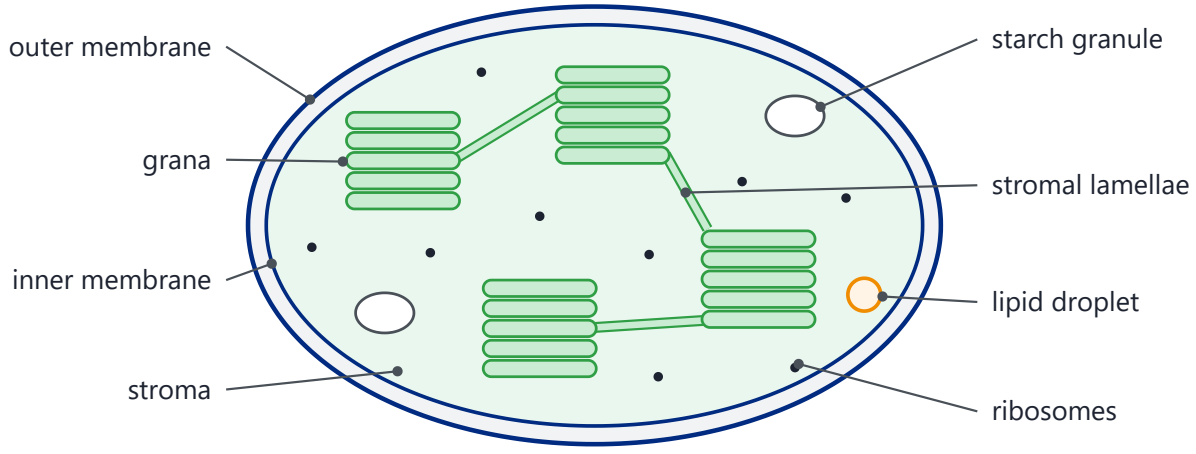
- Calculating the ATP : reduced NADP ratio for a number of turns (WBI14 Jan 2025).
- Phosphate is needed for ATP, nucleotides and phospholipids; nitrate for amino acids and the bases in nucleotides (WBI14 Jun 2023).
- Mineral ions such as phosphate are taken up by **active transport** and travel in the **xylem** (WBI14 Jan 2023).
- Mutualistic fungi (mycorrhizae) supply plants with nitrogen compounds (WBI14 Jun 2024).

SPEC 5.5 Chloroplast structure

WHAT THIS SPEC POINT ASKS

Understand how the structure of a chloroplast suits it to photosynthesis.

The parts and why they help



The stroma is the site of the light-independent reactions, and the thylakoid stacks (grana) are the site of the light-dependent reactions.

Feature	Description	How it helps photosynthesis
Envelope	double membrane (inner and outer)	controls what enters and leaves; inner membrane has transport proteins

Feature	Description	How it helps photosynthesis
Thylakoids	flattened membrane sacs	membranes hold photosystems, electron carriers and ATP synthase: the light-dependent reactions
Grana (singular granum)	stacks of thylakoids	huge membrane **surface area** for pigments and light absorption
Intergranal lamellae	thylakoid membranes joining grana	also carry PSI; link the grana
Thylakoid space (lumen)	inside each thylakoid	small volume, so a **proton gradient** builds quickly for ATP synthesis
Stroma	fluid matrix around the thylakoids	contains **RUBISCO** and the other Calvin cycle enzymes; close to the thylakoids, so ATP and reduced NADP reach it quickly
Starch grains	in the stroma	store the glucose made
Circular DNA and **70S ribosomes**	in the stroma	code for and make some chloroplast proteins (e.g. part of RUBISCO)
Lipid droplets	in the stroma	lipid store (not credited as a "feature" label)

- Chloroplasts are about **2–10 µm** long, so you need an electron microscope to see the grana.
- Not every plant cell has them: root cells and phloem sieve tubes don't.

EXAM TIP

When asked how a chloroplast is adapted, link each structure to a stage: thylakoid membranes with photosystems, electron carriers and ATP synthase for the light-dependent reactions; a lumen in which H^+ accumulates; and a stroma containing RUBISCO and other Calvin cycle enzymes.

WORKED EXAMPLE

Worked example: actual size of a chloroplast

On an electron micrograph at ×8000 a chloroplast measures 58 mm.

Actual size = image size ÷ magnification. Convert first: 58 mm = 58 000 µm.

58 000 ÷ 8000 = 7.25 µm = **7.3 µm** (2 s.f.).

In standard form, keep **one non-zero digit before the decimal point**: 7.3×10^{-3} mm.

MARK-SCHEME WORDING

- Labels that score (beyond stroma and thylakoid): "DNA (loop)", "starch grain", "{envelope / inner membrane / outer membrane}", "grana", "intergranal lamellae", "(70S) ribosome". Lipid droplets are ignored; "linear DNA" and "stoma" are wrong.
- Chloroplast v mitochondrion: **similarities** "both have (circular) DNA", "both have (70S) ribosomes", "both have a double membrane"; **differences** "stroma v matrix", "thylakoids / grana v cristae", "starch grains only in chloroplasts".
- Thylakoid membrane v cell surface membrane: "both have a phospholipid bilayer"; thylakoid "has chlorophyll / photosystems, ATP synthase, electron carriers".
- Why lots of thylakoids: comparative, "**more** light absorbed / **more** chlorophyll / **more** ATP and reduced NADP".

Often asked as: draw and label three features; compare with a mitochondrion; compare membranes; label MCQs; magnification and actual size; why many thylakoids.

COMMON MISTAKE

Compare questions answered with two separate descriptions. Pair each point: "**both** have…", "chloroplasts have… **whereas** mitochondria have…". Separate points are not pieced together.

Describing what the membranes **do** when the question asks about **structure**.

Calling a chloroplast "a cell", or labelling "inner part" instead of a named structure.

Standard form with two digits before the decimal point (73×10^{-4}).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Chloroplasts and mitochondria both have **70S ribosomes and circular DNA**, like bacteria (WBI14 Jun 2021).
- Chloroplast DNA replicates in the light, independently of mitosis (WBI14 Oct 2021).
- Size conversions between nm, μm and mm for ribosomes, thylakoids and starch grains (WBI14 Jan 2022).
- Amyloplasts and vacuoles contain no DNA (WBI14 Jun 2025).

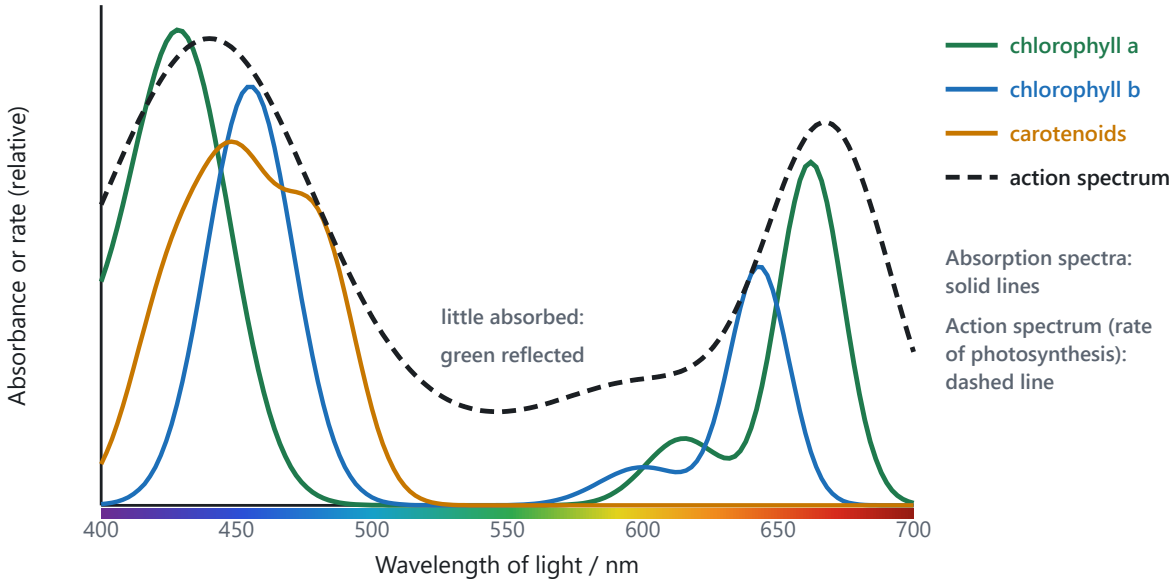
SPEC 5.6 Absorption and action spectra

WHAT THIS SPEC POINT ASKS

Understand what an absorption spectrum and an action spectrum are.

Definitions

- **Absorption spectrum**: how much light **a pigment absorbs** at each wavelength.
- **Action spectrum**: the **rate of photosynthesis** at each wavelength of light.



Sketch of the absorption spectra (solid) and the action spectrum (dashed). Peaks are in blue-violet (about 430–480 nm) and red (about 640–680 nm). The action spectrum follows the combined absorption of all the pigments.

The pigments

Pigment	Colour	Absorbs mainly
Chlorophyll a	blue-green	blue-violet (~430 nm) and red (~660 nm); the reaction-centre pigment
Chlorophyll b	yellow-green	blue (~450 nm) and orange-red (~640 nm)
Carotenoids (β-carotene, xanthophylls)	orange, yellow	blue-violet (400–500 nm)

- Chlorophylls absorb little **green** light: it is **reflected** (or transmitted), so leaves look green.
- The **accessory pigments** (chlorophyll b, carotenoids) absorb wavelengths chlorophyll a absorbs poorly, and pass the energy on to the reaction centre. Having several pigments **widens the range of wavelengths** a plant can use.

Reading the two together

- The action spectrum **closely follows** the combined absorption spectra. That is the evidence that the light absorbed by these pigments is what drives photosynthesis.
- The action spectrum **doesn't fall to zero** in the green: some green light is absorbed by accessory pigments, and light bouncing around inside a thick leaf gets more chances to be absorbed.
- To measure an action spectrum: shine light of one wavelength at a time (coloured filters or LEDs) of the **same intensity** onto an aquatic plant, and measure the rate of oxygen production (as in CP10, 5.8).

MARK-SCHEME WORDING

- Action spectrum: "**rate of photosynthesis** at different wavelengths of light".
- Sketching an action spectrum from an absorption spectrum: "a line that roughly follows the contours of the absorption spectrum" and "does not drop to zero"; the blue peak is higher than the red peak.
- Why several pigments: "so that light can be absorbed at different wavelengths", "so that the rate of photosynthesis is maximised".
- Least absorbed: **green** (around 500–560 nm).
- Ratios of peak heights: give as a ratio (e.g. 1.2 : 1), not a percentage.

Often asked as: define action spectrum; draw it on an absorption graph; why plants have different pigments; read absorbance values; which colour of light boosts growth.

COMMON MISTAKE

Swapping the two definitions. Absorption = **pigment absorbs**; action = **rate of photosynthesis**.

Long answers to a 2-mark question. "To maximise light absorption" is usually the key phrase, not a description of the graph in words.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Seaweeds at different depths**: different seaweeds have different pigment mixtures, so each absorbs the wavelengths that reach its depth (red light penetrates least far). Living at different depths also reduces competition for light (WBI14 Jun 2024).
- Photosynthetic bacteria have absorption peaks at different wavelengths from plants (about 380 nm and 880 nm), so they can use light plants don't (WBI14 Jan 2023).
- Blue and red light give the highest growth rates; amber the lowest (WBI14 Oct 2024).

SPEC 5.7 Chromatography of chloroplast pigments

WHAT THIS SPEC POINT ASKS

Understand that chloroplast pigments can be separated by chromatography and identified by their Rf values.

How it works

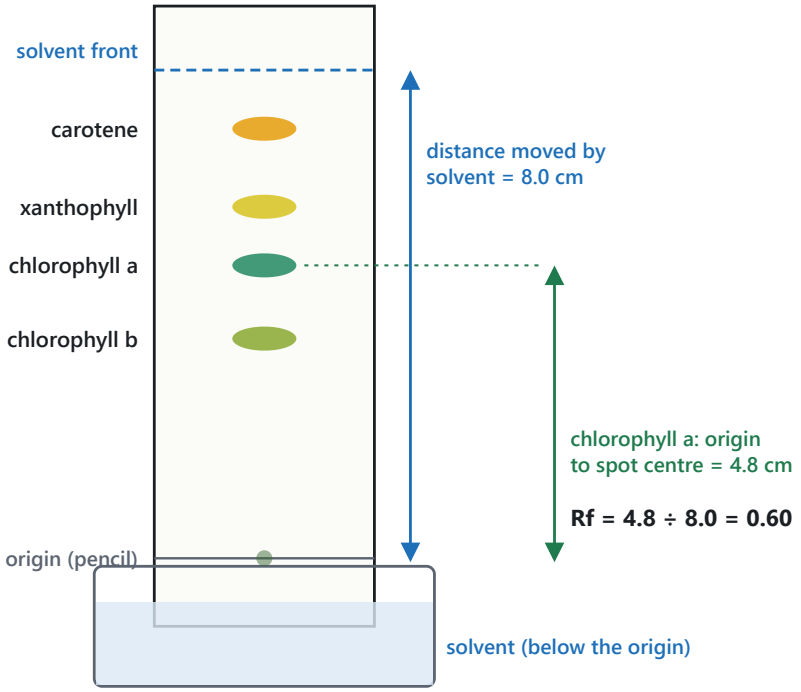
- A **stationary phase** (chromatography paper or a thin-layer plate) and a **mobile phase** (a non-polar solvent).
- Each pigment has its own balance between **solubility in the solvent** and **attraction to the stationary phase**. More soluble pigments are carried further. So the pigments separate into bands.

CORE PRACTICAL

Pigment chromatography: method in brief

1. Grind a **fresh, green** leaf with a little sand and **propanone** (acetone). Grinding breaks cells and membranes; the solvent dissolves the pigments (they are not water-soluble).
2. Draw an **origin line in pencil** about 2 cm from the bottom of the paper or TLC plate. (Ink would dissolve and run.)
3. Use a fine capillary tube to put a **small, concentrated spot** of extract on the origin. Let it dry, then add more on the same spot. Repeat 5–10 times.
4. Stand the paper in a little solvent in a covered tank, with the **solvent level below the origin** (otherwise the pigments dissolve into the solvent pool).
5. Remove it when the solvent is near the top. **Mark the solvent front at once** in pencil, before it evaporates.
6. Measure from the origin to the **centre of each spot** and from the origin to the solvent front. Work out each Rf.

Safety: the solvents are flammable and harmful; no flames, good ventilation or a fume cupboard.



A typical order of bands, top to bottom: carotene, xanthophyll, chlorophyll a, chlorophyll b. The exact Rf values depend on the solvent.

Rf = distance moved by the pigment ÷ distance moved by the solvent front

- Rf has **no units** and is always **between 0 and 1**.
- In one solvent, at one temperature, on one type of paper, each pigment has a **constant Rf**. Compare your Rf with **reference values for the same solvent**, or run known pigments alongside on the same paper in the same tank.

Calculating an Rf value

1. Measure from the pencil line to the solvent front. Suppose this is 10.0 cm.
2. Measure from the pencil line to the centre of the pigment spot. Suppose this is 6.5 cm.
3. Rf = distance moved by the pigment ÷ distance moved by the solvent = 6.5 ÷ 10.0 = 0.65.

An Rf value is a ratio, so it does not depend on how far the solvent ran. It always lies between 0 and 1, and for a given pigment, solvent and paper it is constant, so it can be compared with known values to identify the pigment.

You can run a virtual chromatogram in the interactive section at the bottom of the page and measure the Rf value of each spot.

WORKED EXAMPLE

Worked example: identifying a pigment

The solvent front moved 8.0 cm from the origin. A blue-green spot moved 4.8 cm (to its centre).

Rf = 4.8 ÷ 8.0 = **0.60**.

The reference table for this solvent gives chlorophyll a = 0.60, so the spot is chlorophyll a. Quote Rf to 2 or 3 decimal places.

MARK-SCHEME WORDING

- Rf: "distance moved by pigment ÷ distance moved by solvent front", measured "from the origin" to the "{centre / leading edge}" of the spot.
- Identify: "calculate the Rf value", "compare with Rf values of known pigments **in the same solvent**", or "run known pigments in the same solvent in the same tank".
- Why Rf identifies β-carotene: it is soluble in that solvent, has a specific Rf in that solvent, and is coloured so it can be seen.
- Why two methods give different Rf: different solvents, different solubilities, different stationary phase.
- Technique points: pencil origin, small concentrated spot (dried between applications), solvent below the origin, mark the solvent front straight away.

Often asked as: which technique separates pigments; calculate Rf from a diagram; use Rf to identify a pigment; describe how to measure Rf; the same method applied to antibiotics or proteins.

COMMON MISTAKE

Missing the words "**same solvent**". The examiners say this is the key mark.

Describing how to set up a chromatogram when the question asks how to **identify** a pigment.

Measuring to the top of the paper instead of the solvent front.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The same method separates non-pigments (antibiotics, proteins): the most soluble or smallest molecules travel furthest (WBI14 Jun 2023, Oct 2022).
- Comparing pigment bands from spinach and seaweeds to explain their colours (WBI14 Jun 2024).
- Different solvents extract pigments differently (WBI14 Oct 2021).

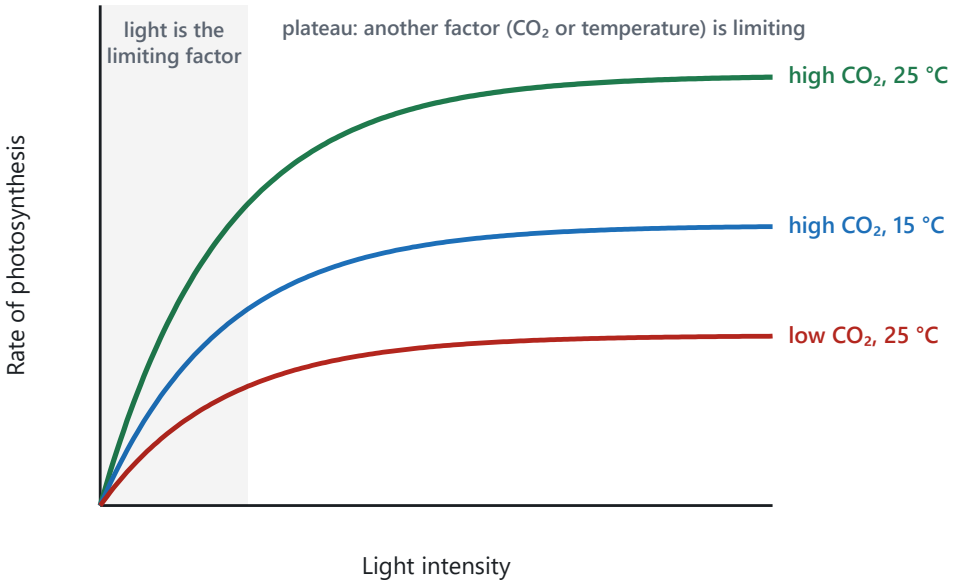
SPEC 5.8 Core Practical 10: factors affecting the rate of photosynthesis

WHAT THIS SPEC POINT ASKS

Investigate how light intensity, light wavelength, temperature and CO_2 availability affect the rate of photosynthesis in an aquatic plant.

Limiting factors

- The rate of photosynthesis is limited by whichever factor is in **shortest supply** (the **limiting factor**): light intensity, CO_2 concentration or temperature.
- Raising the limiting factor increases the rate, until something else becomes limiting and the graph **levels off**.



On the rising part, light is limiting. On the plateau, another factor is limiting: raising CO_2 or temperature lifts the plateau.

- **Light**: provides energy for the light-dependent reactions (exciting electrons, photolysis). More light → more ATP and reduced NADP.
- **CO_2**: the substrate for RUBISCO. More CO_2 → more fixation → more GP.
- **Temperature**: affects the **enzymes** (RUBISCO, ATP synthase). Warmer → more kinetic energy → more enzyme–substrate collisions, up to the optimum; then enzymes denature and the rate falls. Temperature has little effect at low light intensity, because light is limiting there.

Identifying the limiting factor on a graph

1. Find the part of the curve that is still rising. In that region the factor on the x-axis is limiting.
2. Find where the curve levels off. Beyond that point the x-axis factor is no longer limiting.
3. Use the other conditions in the question to name the factor that now limits the rate. If a second curve at a higher CO_2 concentration or temperature levels off higher, that factor was limiting the first curve.

Examiners want the factor that is limiting, not just the factor being changed on the axis.

Practise this on the interactive graph at the bottom of the page, where you set the CO_2 concentration and temperature yourself.

CORE PRACTICAL

Core practical 10: method in brief

1. Cut a 5–10 cm piece of pondweed (*Elodea* or *Cabomba*) underwater and place it, cut end up, in a boiling tube of **sodium hydrogencarbonate** solution (a CO_2 source; use excess so CO_2 isn't limiting).
2. Stand the tube in a **water bath** (or a beaker of water) to keep the temperature constant. Darken the room.
3. Place the lamp (an LED gives off little heat) at a set distance, e.g. 10 cm, measured with a ruler. Leave the plant **5 minutes to acclimatise**.
4. Collect the gas for a set time (e.g. 5 minutes) in a **photosynthometer** (capillary tube + syringe): draw the bubble into the capillary and measure its **length**. Or count bubbles per minute, or use a gas syringe or an oxygen probe.
5. Repeat at least 3 times at each distance and calculate a mean. Then move the lamp to 20, 30, 40 cm…

Changing the other factors:

- **Wavelength**: coloured filters or coloured LEDs at the same distance and intensity.
- **CO_2**: different concentrations of sodium hydrogencarbonate, other factors fixed.
- **Temperature**: water baths at e.g. 10, 20, 30, 40 °C, with the light fixed.

Variables: IV = the factor you change. DV = volume of oxygen per unit time. Control: the other three factors, the same plant (same mass and length), acclimatisation time, collection time.

Processing the results

- Volume of gas in a capillary = $\pi r^2 l$ (r = radius of the bore, l = length of the bubble).
- Rate = volume ÷ time.
- Light intensity is proportional to $1/d^2$ (d = distance from the lamp): the **inverse square law**. Doubling the distance cuts intensity to a quarter. Plot rate against $1/d^2$ for a direct measure of intensity.

WORKED EXAMPLE

Worked example: rate from a photosynthometer

Capillary bore radius r = 0.5 mm. Collection time 5 min.

Distance / cm	$1/d^2$ / cm^{-2}	Bubble length / mm (mean of 3)	Volume / mm^3	Rate / $mm^3\ min^{-1}$
10	0.0100	24.0	18.8	3.77
20	0.0025	8.0	6.3	1.26
40	0.000625	2.2	1.7	0.35

At 10 cm: $V = \pi \times 0.5^2 \times 24.0 = 18.85\ mm^3$ (18.8 to 3 s.f.); rate = 18.85 ÷ 5 = **3.77 $mm^3\ min^{-1}$**.

Moving from 20 cm to 10 cm makes the light **4 times** as intense (1/100 ÷ 1/400) and the rate rises 3.0 times. Rate is not exactly proportional, so another factor is starting to limit at high intensity.

Errors and improvements

Problem	Why it matters	Fix
Lamp heats the water	temperature changes as well as light	LED, heat shield (a tank of water), or water bath
Bubbles vary in size	counting bubbles is inaccurate	measure volume in a photosynthometer or gas syringe
Some O_2 dissolves in the water; some is used in respiration	you measure **net** O_2 release, which underestimates the true rate	treat results as comparative; same conditions throughout
Gas is not pure O_2	some N_2 and CO_2 diffuse in	still fine for comparing
Background light	uncontrolled light intensity	darkened room
CO_2 used up over time	CO_2 becomes limiting	excess sodium hydrogencarbonate; fresh solution

MARK-SCHEME WORDING

- Units for rate: e.g. "mg CO_2 **used** $mm^{-2}\ hr^{-1}$" (per area per time), or volume of O_2 per unit time.

- Tangent for a rate from a curve: "tangent correctly drawn, touching the curve at the stated time", answer to a sensible number of decimal places.
- Temperature effect: "more collisions", "RUBISCO works faster", "ATP synthase works faster".
- Light no longer limiting: "light (intensity) is not rate limiting" once extra pigment widens the range of wavelengths absorbed.

Often asked as: the method itself hasn't been set in WBI14 papers 2020–2026, but its ideas have: rate against temperature for two species, a tangent on an O_2 curve, wavelength and biomass, rate units. Learn the method anyway; the spec names it.

COMMON MISTAKE

Saying temperature affects the light-dependent reactions "by giving more light". It affects **enzymes**.

Reading two lines' crossing point carelessly. Read graph values to the nearest small square.

A tangent that doesn't touch the curve at the time asked.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **C3 and C4 plants**: C4 plants (e.g. maize) have a higher optimum temperature (about 34–36 °C, against 26–28 °C for C3 plants) (WBI14 Jan 2024).
- Different species have different optimum temperatures for photosynthesis, which helps explain where they grow (WBI14 Jan 2022).
- Temperature, RUBISCO and ATP synthase as the explanation of a rate-against-temperature curve (WBI14 Jan 2026).

Quick checks

QUICK CHECK

Why do leaves appear green?

- Chlorophyll absorbs green light most strongly.
- Carotenoids absorb blue and red light and reflect yellow light.
- **Chlorophyll absorbs red and blue-violet light and reflects most of the green light.**
- The stroma contains a green pigment that absorbs all wavelengths.

Chlorophyll a absorbs most strongly in the blue-violet and red regions, so mostly green light is reflected to the eye. The commonest wrong answer says chlorophyll absorbs green light, which is the reverse of what happens.

QUICK CHECK

Which statement about ATP is correct?

- **Phosphorylation of ADP requires energy, and hydrolysis of ATP releases energy.**
- Hydrolysis of ADP releases energy, and phosphorylation of ATP requires energy.
- Phosphorylation of ADP releases energy that drives the Calvin cycle.
- ATP contains deoxyribose and three phosphate groups.

Adding Pi to ADP (phosphorylation) needs an energy input; removing it by hydrolysis releases energy for cell processes. The commonest wrong answer swaps the two processes. ATP contains ribose, not deoxyribose.

QUICK CHECK

In non-cyclic photophosphorylation, where do the electrons that replace those lost by photosystem I come from?

- From the photolysis of water.
- From reduced NADP made in the stroma.
- From cyclic flow, which returns them to P700.
- From photosystem II, passed along the electron carriers.

Electrons from photosystem II travel along the carrier chain to photosystem I. Photolysis replaces the electrons lost by photosystem II, not those lost by photosystem I, which is the commonest wrong answer.

QUICK CHECK

The Calvin cycle fixes 6 CO_2 to form one glucose. How many ATP and reduced NADP are used in total?

- 12 ATP and 12 reduced NADP
- 18 ATP and 12 reduced NADP
- 12 ATP and 18 reduced NADP
- 6 ATP and 6 reduced NADP

Each of the 12 GP needs one ATP and one reduced NADP to be reduced to GALP, giving 12 of each. Regenerating 6 RuBP needs another 6 ATP, so the total is 18 ATP and 12 reduced NADP. The commonest wrong answer, 12 of each, leaves out the ATP used to regenerate RuBP.

QUICK CHECK

A pigment travels 4.0 cm while the solvent front travels 10.0 cm. What is its Rf value?

- 0.25
- 2.5
- 6.0
- 0.40 ✓

Rf = distance moved by the pigment ÷ distance moved by the solvent = 4.0 ÷ 10.0 = 0.40. The commonest wrong answer, 2.5, comes from dividing the other way round, which gives a value above 1 that is impossible for an Rf.

QUICK CHECK

A plant in bright light is suddenly placed in dim light. What happens to the concentrations of GP and RuBP in its chloroplasts?

- GP falls and RuBP rises.
- GP rises and RuBP falls. ✓
- Both rise.
- Both stay the same.

With less ATP and reduced NADP, GP cannot be reduced to GALP and accumulates, while RuBP is still used for fixation but cannot be regenerated, so it falls. GP falling and RuBP rising is what happens when the carbon dioxide concentration falls, which is the commonest confusion.

QUICK CHECK

What does cyclic photophosphorylation produce?

- Oxygen and ATP
- Reduced NADP only
- ATP only
- ATP, reduced NADP and oxygen

Cyclic flow uses photosystem I only, and the electrons return to P700, so hydrogen ions are pumped and ATP is made, but water is not split and NADP is not reduced. The commonest wrong answer lists the products of non-cyclic flow.